

Co 2 Assessment tool-1

Q1) Healthcare Analytics:

Patient	Age	BMI	Blood Pressure	Severity
P1	24	19.8	110	Low
P2	32	24.1	Missing	Medium
P3	41	28.1	Missing	High
P4	48	30.1	Missing	High
P5	56	33.1	Missing	Critical
P6	65	36.1	Missing	Critical
P7	38	26.1	Missing	Medium
P8	29	22.1	Missing	Low

R program:

```
library(ggplot2)
```

```
data <- data.frame(
```

```

Patient = c("P1","P2","P3","P4","P5","P6","P7","P8"),
Age = c(24,32,41,48,56,65,38,29),
BMI = c(19.8,24.1,28.1,30.1,33.1,36.1,26.1,22.1),
BloodPressure = c(110,NA,NA,NA,NA,NA,NA,NA),
Severity = factor(
c("Low","Medium","High","High","Critical","Critical",
"Medium","Low"),
  levels = c("Low","Medium","High","Critical")
)
)
ggplot(data, aes(x = Age, y = BloodPressure,
  color = Severity, size = BMI)) +
  geom_point(alpha = 0.8) +
  scale_color_manual(
    values = c(
      "Low" = "green",
      "Medium" = "yellow",
      "High" = "orange",
      "Critical" = "red"
    )
  )

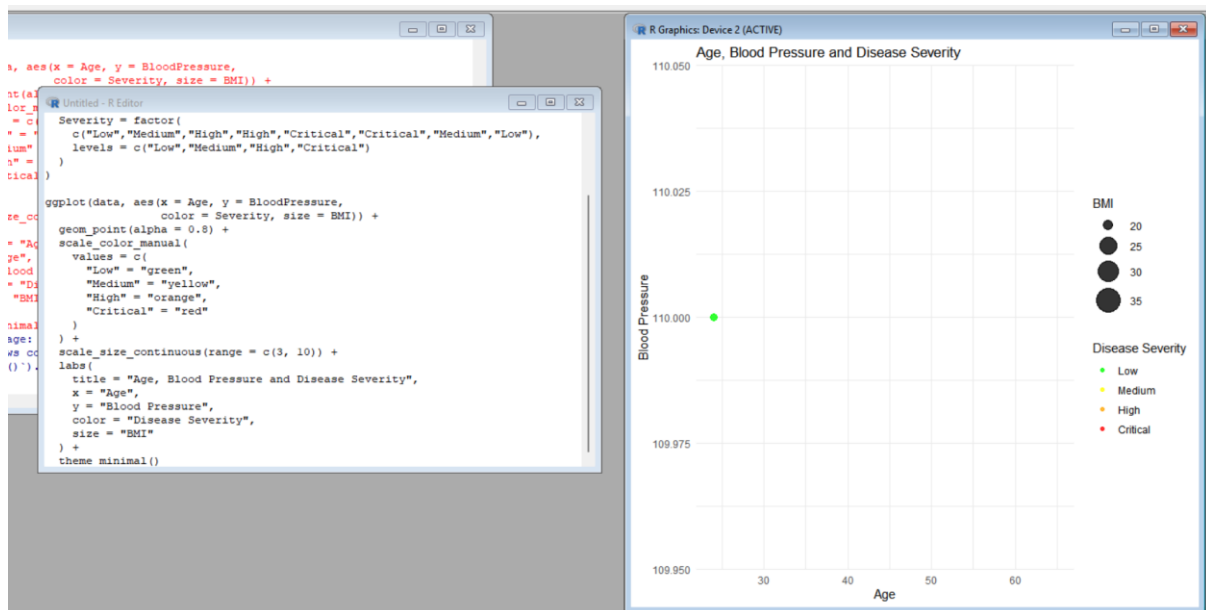
```

```

) +
scale_size_continuous(range = c(3, 10)) +
labs(
  title = "Age, Blood Pressure and Disease Severity",
  x = "Age",
  y = "Blood Pressure",
  color = "Disease Severity",
  size = "BMI"
) +
theme_minimal()

```

Output:



Justification

- Age on X-axis makes age-related health-risk patterns easy to observe.
- Blood Pressure on Y-axis allows abnormal pressure levels to be identified.
- Disease Severity by colour provides an immediate risk classification from Low to Critical.
- BMI by point size allows higher BMI values to be visually distinguished.
- A green → yellow → orange → red scale communicates increasing health risk intuitively.
- Larger points represent higher BMI, making multiple health-risk indicators visible simultaneously.

Q2) Climate Science:

Month	Radiation	Temperature	Rainfall
Jan	180	20	25
Feb	200	23	18
Mar	240	29	12

Month	Radiation	Temperature	Rainfall
Apr	280	35	8
May	320	39	5
Jun	300	36	65
Jul	260	32	150
Aug	250	31	170
Sep	240	30	120
Oct	220	28	85
Nov	190	24	45
Dec	170	21	30

i) R code — Cartesian Temperature Line Chart

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Month = factor(
```

```
    c("Jan","Feb","Mar","Apr","May","Jun",
```

```
      "Jul","Aug","Sep","Oct","Nov","Dec"),
```

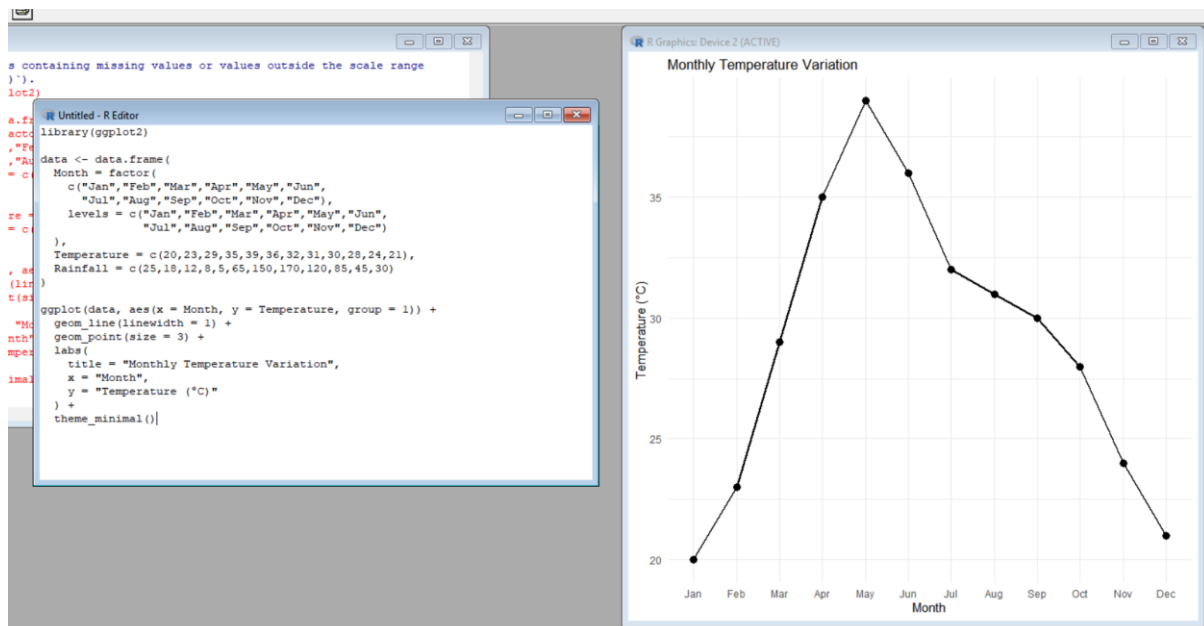
```

    levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
               "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
  ),
  Temperature =
c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),
  Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)
)

ggplot(data, aes(x = Month, y = Temperature, group =
1)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Monthly Temperature Variation",
    x = "Month",
    y = "Temperature (°C)"
  ) +
  theme_minimal()

```

Output:



ii) R code- polar rainfall chart

library(ggplot2)

data <- data.frame(

Month = factor(

c("Jan", "Feb", "Mar", "Apr", "May", "Jun",

"Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),

levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",

"Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

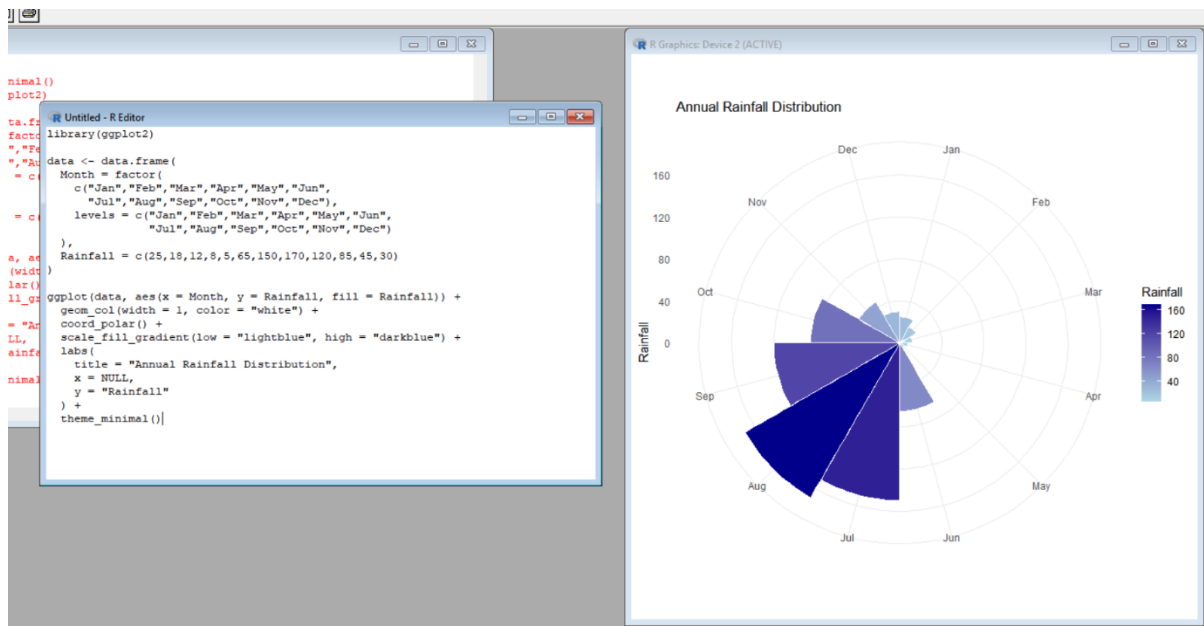
),

Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)

)

```
ggplot(data, aes(x = Month, y = Rainfall, fill =  
Rainfall)) +  
  geom_col(width = 1, color = "white") +  
  coord_polar() +  
  scale_fill_gradient(low = "lightblue", high =  
"darkblue") +  
  labs(  
    title = "Annual Rainfall Distribution",  
    x = NULL,  
    y = "Rainfall"  
  ) +  
  theme_minimal()
```

Output:



Comparison

Cartesian coordinate system — Advantages

- Easy to read exact temperature values.
- Clearly shows increasing and decreasing trends.
- Best for comparing monthly temperature changes.
- Suitable for time-series analysis.

Cartesian — Limitations

- Seasonal/cyclic patterns are less visually circular.
- Annual repetition is not emphasized.

Polar coordinate system — Advantages

- Excellent for displaying cyclic or seasonal data.
- Months naturally form a complete annual cycle.

- Makes rainfall concentration during particular seasons easy to identify.

Polar — Limitations

- Exact values are harder to compare.
- Small differences between neighbouring months can be difficult to judge.
- Less suitable when precise numerical comparison is required.

Q3) Business Analytics & Marketing

Product	Sales (₹ Lakhs)	Profit %	Market Share %
A	210	18	24
B	185	14	20
C	260	25	28
D	145	12	15
E	120	10	13

R code:

```
library(ggplot2)
```

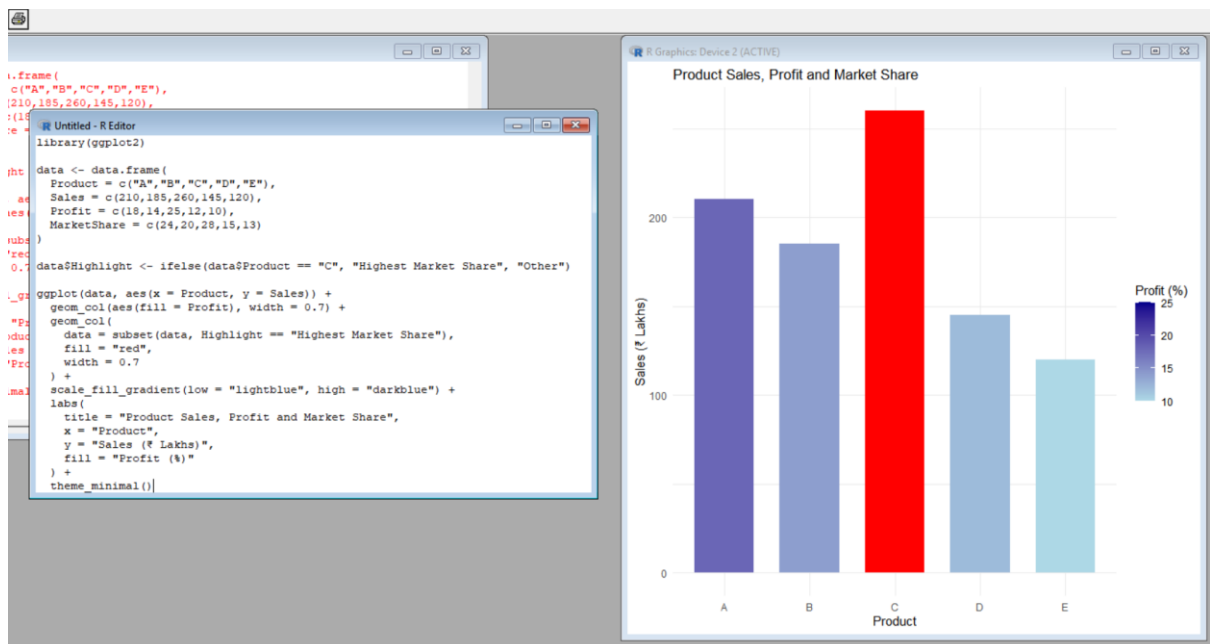
```
data <- data.frame(  
  Product = c("A","B","C","D","E"),  
  Sales = c(210,185,260,145,120),  
  Profit = c(18,14,25,12,10),  
  MarketShare = c(24,20,28,15,13)  
)
```

```
data$Highlight <- ifelse(data$Product == "C", "Highest  
Market Share", "Other")
```

```
ggplot(data, aes(x = Product, y = Sales)) +  
  geom_col(aes(fill = Profit), width = 0.7) +  
  geom_col(  
    data = subset(data, Highlight == "Highest Market  
Share"),  
    fill = "red",  
    width = 0.7  
  ) +  
  scale_fill_gradient(low = "lightblue", high =  
"darkblue") +
```

```
labs(  
  title = "Product Sales, Profit and Market Share",  
  x = "Product",  
  y = "Sales (₹ Lakhs)",  
  fill = "Profit (%)"  
) +  
theme_minimal()
```

Output:



Justification

- Bar height represents Sales, making product sales easy to compare.

- Sequential colour represents Profit %, where darker colour indicates greater profit.
- Product C is highlighted in red because it has the highest market share of 28%.
- The sequential colour scale helps managers quickly identify products with stronger profitability.
- The contrasting highlight draws attention to the most important market-share product.
- This combination supports quick sales, profitability, and market-position decisions.

Q4) Transportation Engineering:

The direction list contains:

- North
- North-East
- East
- South-East
- South
- South-West
- West
- North-West

But the supplied PDF only clearly provides these traffic values:

- North = 980
- East = 1240
- South = 1420
- West = 860
- North-West = 720

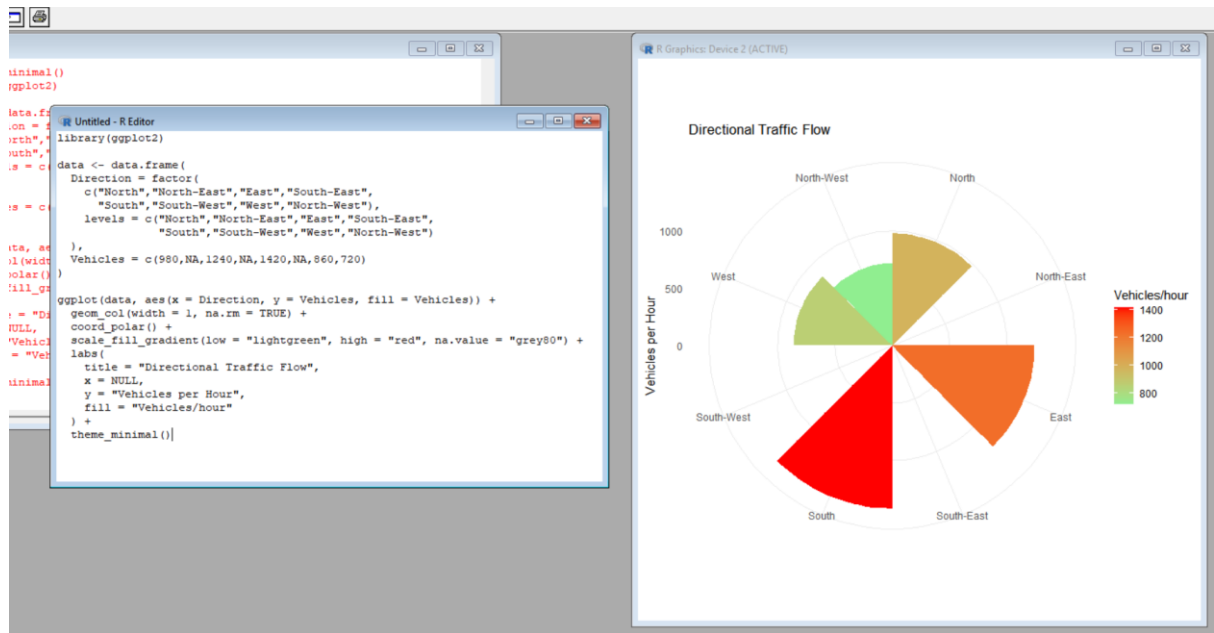
R code:

```
library(ggplot2)
```

```
data <- data.frame(  
  Direction = factor(  
    c("North","North-East","East","South-East",  
      "South","South-West","West","North-West"),  
    levels = c("North","North-East","East","South-  
East",  
               "South","South-West","West","North-  
West")  
  ),  
  Vehicles = c(980,NA,1240,NA,1420,NA,860,720)  
)
```

```
ggplot(data, aes(x = Direction, y = Vehicles, fill =  
Vehicles)) +  
  geom_col(width = 1, na.rm = TRUE) +  
  coord_polar() +  
  scale_fill_gradient(low = "lightgreen", high = "red",  
na.value = "grey80") +  
  labs(  
    title = "Directional Traffic Flow",  
    x = NULL,  
    y = "Vehicles per Hour",  
    fill = "Vehicles/hour"  
  ) +  
  theme_minimal()
```

Output:



Justification

- A polar chart is appropriate because traffic direction is inherently angular.
- Each direction can be represented around a 360° circular coordinate system.
- Traffic volume is represented by radial magnitude.
- High-density traffic directions can be identified immediately.
- A Cartesian bar chart treats directions as unrelated categories, whereas a polar chart preserves their spatial/directional relationship.
- This makes the visualization more useful for transportation engineering and directional traffic analysis.

Q4) Environmental Sustainability Dashboard:

Location	AQI	Green Cover %	Emission	Population (Lakhs)
A	48	45	1.8	9
B	82	32	3.5	15
C	135	21	5.6	22
D	165	18	7.4	28
E	72	38	2.9	12
F	95	30	4.2	18

R code:

```
library(ggplot2)
```

```
library(gridExtra)
```

```
data <- data.frame(
```

```
  Location = c("A","B","C","D","E","F"),
```

```
  AQI = c(48,82,135,165,72,95),
```

```
  GreenCover = c(45,32,21,18,38,30),
```

```
  Emission = c(1.8,3.5,5.6,7.4,2.9,4.2),
```

```
Population = c(9,15,22,28,12,18)  
)
```

```
data$AQI_Category <- cut(  
  data$AQI,  
  breaks = c(-Inf,50,100,150,200,Inf),  
  labels =  
  c("Good","Satisfactory","Moderate","Poor","Very  
  Poor")  
)
```

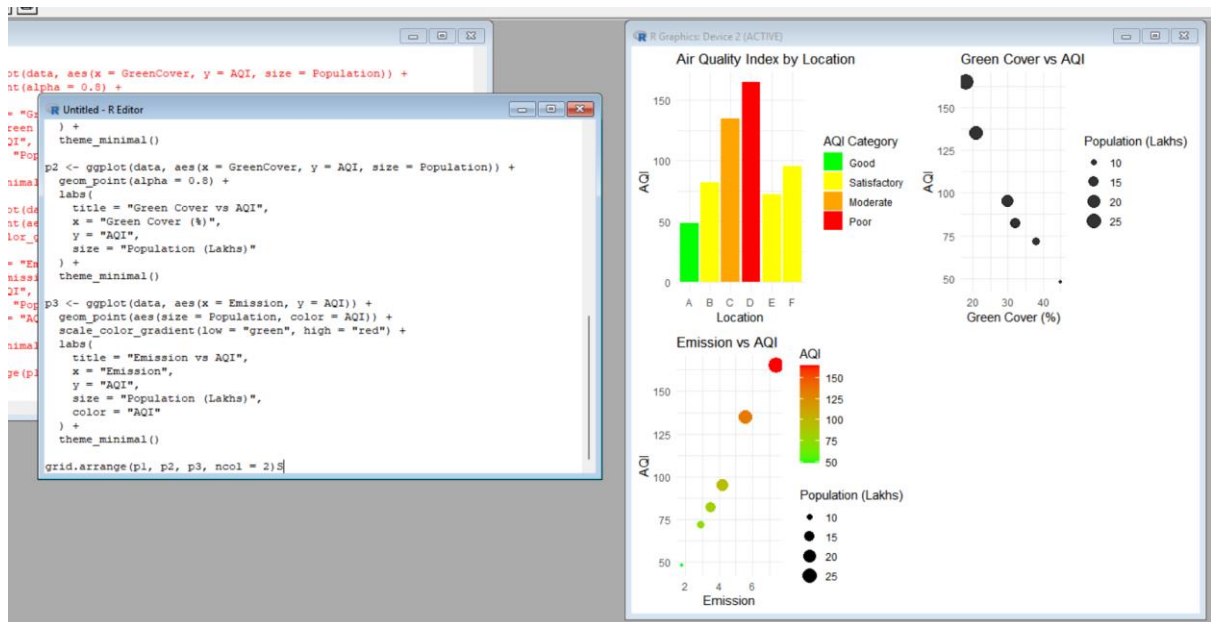
```
p1 <- ggplot(data, aes(x = Location, y = AQI, fill =  
  AQI_Category)) +  
  geom_col() +  
  scale_fill_manual(  
    values = c(  
      "Good" = "green",  
      "Satisfactory" = "yellow",  
      "Moderate" = "orange",  
      "Poor" = "red",  
      "Very Poor" = "darkred"
```

```
)  
) +  
labs(  
  title = "Air Quality Index by Location",  
  x = "Location",  
  y = "AQI",  
  fill = "AQI Category"  
) +  
theme_minimal()
```

```
p2 <- ggplot(data, aes(x = GreenCover, y = AQI, size  
= Population)) +  
  geom_point(alpha = 0.8) +  
  labs(  
    title = "Green Cover vs AQI",  
    x = "Green Cover (%)",  
    y = "AQI",  
    size = "Population (Lakhs)"  
  ) +  
  theme_minimal()
```

```
p3 <- ggplot(data, aes(x = Emission, y = AQI)) +  
  geom_point(aes(size = Population, color = AQI)) +  
  scale_color_gradient(low = "green", high = "red") +  
  labs(  
    title = "Emission vs AQI",  
    x = "Emission",  
    y = "AQI",  
    size = "Population (Lakhs)",  
    color = "AQI"  
  ) +  
  theme_minimal()  
  
grid.arrange(p1, p2, p3, ncol = 2)
```

Output:



Justification

1. AQI bar chart

- AQI is represented using bar height because locations need direct comparison.
- A categorical colour scale communicates AQI severity clearly.
- Green represents better air quality, while red represents poorer air quality.

2. Green Cover vs AQI

- Green Cover is placed on the X-axis and AQI on the Y-axis.
- This helps identify the relationship between vegetation and air quality.

- Population is represented using point size.

3. Emission vs AQI

- Emission is placed on the X-axis and AQI on the Y-axis.
- This helps identify whether higher emissions are associated with poorer air quality.
- Population size provides an additional demographic dimension.

4. Continuous scales

- AQI, green cover, emission and population are quantitative variables, so continuous scales are appropriate.
- Continuous axes preserve numerical differences without unnecessary categorization.

5. Dashboard design

- Multiple coordinated plots allow environmental factors to be examined together.
- The dashboard supports comparison of air quality, vegetation, emissions and population.